Lie Groups — Lecture 3

November 1, 2026

1 Reminder

Definition 1.1 (Tangent space)**.** If (X, A) is a smooth manifold and $p \in X$ is some point, then the tangent space of p is defined as,

$$T_pX = \{\gamma : (-\delta, \delta) \to X \mid \gamma(0) = p, \gamma \in C^\infty(X), \delta \in \mathbb{R}^+\}/\sim$$

Definition 1.2 (exponential map)**.** If G is a Lie group then define $\exp : L(G) \to G$ as the map that for any $v \in L(G)$, θ_v be the corresponding 1-parameter subgroup of G, then $\exp tv = \theta_v(t)$.

TODO: Maybe insert here a graph of $\mathbb{R}^2$ and an embedding of T^2, showing how would $\exp(tv)$ would look for a given $v \in S^1$.

2 Submanifolds

We intend to characterize the subspace of a Lie group, in order to do so we will discuss a subspace retaining the properties of manifold, as the characterization of a subgroup is well defined. The general idea if to define submanifold as the restriction of a manifold to a subspace such that its smooth property, as show up as smoothness of charts, would not be affected.

Definition 2.1 (Submanifold)**.** Let M be a smooth manifold modeled on V, and let $N \subseteq M$ be a subset and $W \subseteq V$ be a subspace. N is a **submanifold** modeled on W if for any $x \in N$ there exists a chart compatible with M, $\varphi_\alpha : M_\alpha \to \Omega_\alpha$ where $M_\alpha \subseteq M, \Omega_\alpha \subseteq V$ are open, that restricts to a bijection $M_\alpha \cap N \to \Omega_\alpha \cap W$.

Remark. The requirement for a compatible chart is crucial, as the given atlas might not contain a single chart matching M_α, instead being equivalent to an atlas that does.

Our first task is to show that a submanifold, in the sense of a manifold subset retaining smoothness properties, is indeed a smooth manifold on its own. We first need an actual atlas to show that, that leads us to,

Proposition 2.2. *Given a manifold M and a submanifold N, the bijections $\{M_\alpha \cap N \to \Omega_\alpha \cap W\}$ are an atlas on N. Furthermore, equivalent atlases on M yield equivalent atlases on N thus N is a smooth manifold modeled on W.*

Proof. Left as an exercise. □

We can now describe the structure of a submanifold as a manifold,

Proposition 2.3. *If N is a submanifold of M, then N can be given differential structure as a manifold so that:*

1. *The inclusion of N in M is smooth*
2. *$P \to N$ is smooth if and only if the composition $P \to N \to M$ is smooth*

Proof. Let $f : N \to M$ be the identity embedding, and let $\varphi : U \to V$ be a chart of M, where $U \subseteq M, V \subseteq \mathbb{R}^k$ are open, and assume $N \cap U \neq \emptyset$. Then φ can be restricted to $\psi : N \cap U \to V \cap W$ a bijection and $\varphi \circ f \circ \psi$ is smooth by our

definition of smoothness and as a restriction.

The second part is similar and is left as an exercise. □

Corollary 2.4. $T(N)$ is embedded in $T(M)$.

Recall that if M_1, M_2 are smooth manifolds, $M_1 \times M_2$ can be given a smooth manifold structure, this holds for submanifolds as well.

Exercise 2.1. Show that if N_1, N_2 are manifolds of M_1, M_2 then $N_1 \times N_2$ is a submanifold of $M_1 \times M_2$, and its differential structure as a submanifold is the same as of the product of the submanifolds as smooth manifolds.

As of now, there are two distinct "substructures" we consider of a Lie group, that is the subgroup & the submanifold. The question that naturally arises is whether there is a connection between these two substructures.

Proposition 2.5. *If G is a Lie group and H is both a submanifold and a subgroup then H is a Lie group.*

Proof. Follows from the last exercise and proposition for the maps $\mu : G \times G \to G$ and $\mu : H \times H \to H$ and $i : G \to G$ to $i : H \to H$. To elaborate we defined the submanifold differential structure as such that preserves smooth maps, then the group operations (that are of course smooth) are being preserved. □

So these two structure do collide when in Lie groups.

3 Closed subgroups

While subgroups are a natural way to understand groups, and are easy to construct, submanifolds are somewhat more difficult object to handle, is there a way to construct submanifolds from subgroups? We will show that in a special case of subgroups the answer is that it is.

Definition 3.1 (Closed subgroup)**.** A subgroup H of a Lie group G is closed if it is closed as a subset of the topology G.

To understand the case of a non-closed subgroup:

Example 3.1. Let $M = T^2 = \mathbb{R}^2/\mathbb{Z}^2$ the 2-torus, and take $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ some irrational number. Define $H = \{(t, \alpha t) \in \mathbb{R}^2 \mid t \in \mathbb{R}\}/\mathbb{Z}^2$, then $H \leq M$ a subgroup, and a dense set of $\mathbb{R}^2/\mathbb{Z}^2$, in particular it isn't closed.

It is also clear that H does not assume a manifold structure, as any small enough open neighborhood of any point $\in H$ is not connected as $H \cap (\mathbb{Q}/\mathbb{Z})^2 = \{0\}$.

In the other hand,

Example 3.2. If $\alpha \in \mathbb{Q}$ then there must be $z \in \mathbb{Z} \setminus \{0\}$ such that $\alpha z \in \mathbb{Z}$, meaning that H is a closed path in the topological space M. In particular H is indeed a 1 dimensional manifold.

This result is no coincidence, it leads us to,

Theorem 3.2 (Closed subgroup as submanifold)**.** *A closed subgroup H of a Lie group G is a submanifold as well.*

The proof of the theorem is given by the next three lemmas:

Lemma 3.3. *Suppose $L(G) = T_eG$ has a norm $\|\cdot\|$, and suppose that $\{h_n\}_{n=1}^{\infty} \subseteq L(G) \setminus \{0\}$ is a sequence of points such that,*

$$\forall n \in \mathbb{N},\ \exp h_n \in H, \qquad h_n \xrightarrow{n\to\infty} 0, \qquad \frac{1}{\|h_n\|} h_n \to v \in L(G)$$

Then $\exp(tv) \in H$ *for all* $t \in \mathbb{R}$.

In other words, a closed subgroup is closed under exponent, in particular under 1-parameter subgroups generated by $v \in H$.

Proof. Suppose $t \in \mathbb{R}$, then,

$$\frac{t}{\|h_n\|} \to tv$$

and $\|h_n\| \to 0$. Let $\{m_n\} \subseteq \mathbb{N}$ be a sequence such that $m_n\|h_n\| \xrightarrow{n\to\infty} t$, such sequence can be obtained directly by limit definition. Then,

$$\lim_{n\to\infty} \exp m_n h_n = \lim_{n\to\infty} \exp \frac{m_n\|h_n\|}{\|h_n\|} h_n = \exp tv$$

But $\exp m_n h_n = (\exp h_n)^{m_n} \in H$ as a subgroup, and $tv \in H$ follows from closure. □

Lemma 3.4. *Let $W \subseteq L(G)$ be a set of all tv for $t \in \mathbb{R}$ and $v \in L(G)$ of the last lemma. Then $\exp W \subseteq H$ and W is a vector subspace of $L(G)$.*

Proof. If $v \in L(G)$ as in the last lemma, then $\exp tv \in H$, W consists of such values therefore $\exp W \subseteq H$.

It is clear that if $w \in W$ then $tw \in W$ for all $t \in \mathbb{R}$, and it remain to show that if $w_1, w_2 \in W$ then $w_1 + w_2 \in W$ as well. □

Lemma 3.5. $\exp W$ *is a neighborhood of $e \in H$.*

Proof. TODO □

Proof of the theorem. By the last lemma $\exp W$ is a neighborhood of e in H, thus $\exp|_W$ is a chart of a neighborhood of e. This chart can be translated to any other point $\in H$ by multiplication. □

Example 3.3. $O(n) \subseteq \mathrm{GL}_n(\mathbb{R})$ the subspace of orthogonal invertible matrices is a closed subgroup (TODO proof) and so a submanifold of a Lie group thus its is a Lie group.

Example 3.4. $U(n) \subseteq \mathrm{GL}_n(\mathbb{C})$ the subspace of unitary complex matrices is also a closed subset (TODO proof) thus also a Lie group.

TODO: Should we copy the third example?

Example 3.5. If G is a compact Lie group and H is a closed connected Abelian subgroup, then H is a torus.

Proposition 3.6. *A closed connected subgroup H of a Lie group G is determined by $L(G)$.*

Proof. TODO something with 2.17 which means that a homomorphism is determined by its projection to the tangent space. □

4 Quotient spaces

Definition 4.1 (Subgroup & Cosets)**.** Suppose G is a Lie group and H is a closed subgroup, then the **quotient space** G/H is the set of (left) cosets $\{gH \mid g \in G\}$. The map $\pi : G \to G/H$ given by $g \mapsto gH$ is the projection, and give G/H the quotient topology.

Proposition 4.2. *G/H is Hausdorff.*

Proof. Suppose that $xH, yH \in G/H$ are different points, conversely $x, y \in G$ such that $x \neq y$ and $xH \neq yH$. G is Hausdorff then there are $x \in U, y \in V$ open neighborhoods such that $U \cap V = \emptyset$. $e \in x^{-1}U, y^{-1}V$ then there is a chart $\varphi : W \to \mathbb{R}^k$ for $W \subseteq x^{-1}U \cap y^{-1}V$ up to compatibility with G. By translating the chart we can make V, U arbitrarily small (taking $\varphi^{-1}(B(0,\varepsilon))$), such that $\varepsilon < \mathrm{dist}(\varphi(xW \cap xH, yW \cap yH))$ if it is defined (otherwise the claim holds). □

Together with the fact that a quotient topology assumes 2nd countability of a topology, we get,

Proposition 4.3. *G/H can be given a smooth manifold structure, such that π is smooth and any $f : G/H \to M$ is smooth if and only if there exists $F : G \to M$ smooth with $f = F \circ \pi$.*

Proof. TODO □

Definition 4.4 (Bundle)**.** TODO

Proposition 4.5. *$H \to G \to G/H$ is a bundle.*